

Psychotic Symptoms in Schizophrenia Are Shaped by Life Factors More Than Expected

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The symptoms you'd expect to be "hardwired" may be the most shaped by life circumstances.

Hallucinations and delusions are often thought of as the most biologically fixed features of psychosis. A new study from Karolinska Institutet suggests this may not entirely be true: these "positive" symptoms appear more sensitive to a person's life circumstances and behaviour than "negative" symptoms such as social withdrawal and diminished emotion.

The study, by Silpa Soni Nallacheruvu in the Master's Programme in Biostatistics and Data Science, used data from the Swedish Twin Registry to examine how substance use, genetic vulnerability, and early-life adversity relate to psychotic symptoms in adolescence. Rather than testing each factor alone, the analysis modeled how they combine — comparing traditional statistical models with a deep learning approach built to capture those interactions.

The two symptom types behaved differently. Positive symptoms were most strongly associated with combinations of risk factors, such as substance use together with childhood adversity or genetic vulnerability. Negative symptoms followed a more stable pattern, shaped more by genetic and individual factors than by how circumstances combined. No single factor stood out; the alignment of several tracked most closely with severity.

"What seems to matter for the more visible symptoms is how factors come together, not just whether they're present," says Nallacheruvu. "Two people with similar exposures can end up in very different places."

The distinction may carry clinical weight. If acute symptoms are more responsive to circumstance, they may also be more open to early, behaviorally focused support, while more stable symptoms may call for longer-term approaches.

The findings are based on observational data and describe associations rather than proven causes. Even so, they suggest that risk is not fixed: the same factor may carry different weight depending on context.

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Statement on the use of generative AI:

Generative AI tools were used to assist in refining the language and structure of this press release. The content and conclusions are the author's own.